Edgar J. Andrade-Lotero, Ph.D.

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PROFESSIONAL SUMMARY

Academic and researcher with experience spanning logic, cognitive science, and artificial intelligence. Expertise in behavioral experiments and computational modeling of group decision-making and collective intelligence. Research collaborations with leading laboratories at Indiana University, UC Davis, and Cornell University. Experienced in supervising undergraduate and graduate students and teaching mathematics, artificial intelligence, and philosophy of language.

EXPERIENCE

Postgraduate Researcher 2025 – Present
University of California, Davis, USA *Department of Communication*

Expert Consultant 2024 – 2025
Turing.com

Associate Professor 2019 – Present
Universidad del Rosario, Bogotá, Colombia *School of Sciences and Engineering*

Associate Professor 2010 – 2019
Universidad del Rosario, Bogotá, Colombia *Faculty of Natural Sciences and Mathematics*

Auxiliary Professor July 2006 – 2010
Universidad del Rosario, Bogotá, Colombia *School of Social Sciences*

Lecturer February 2006 – June 2006
Universidad Sergio Arboleda, Bogotá, Colombia *School of Mathematics*

Lecturer February 2004 – June 2004
Universidad Sergio Arboleda, Bogotá, Colombia *School of Mathematics*

Lecturer 2003 – 2004
Universidad Nacional de Colombia, Bogotá, Colombia *Department of Mathematics*

EDUCATION

University of Amsterdam
Ph.D. in Logic. Thesis: Models of Language: Towards a practice-based account of information in natural language. 2012

University of Amsterdam
M.Sc. in Logic. Thesis: Meaning and Form in Event Calculus. 2006

Universidad Nacional de Colombia
B.A. in Mathematics. 2002

PUBLICATIONS: RESEARCH ARTICLES

- D. Yela, G. Obando, A. Pantoja, **E. J. Andrade-Lotero**, and J. Barreiro-Gomez, "Finding the delay that maximizes the convergence rate of a class of LTI systems," Journal of the Franklin Institute, vol. 363, no. 15, p. 109012, 2026.
- **E. J. Andrade-Lotero**, C. Álvarez, and G. Obando, "A paradox in Brian Arthur's solution to his El Farol Bar problem," Advances in Complex Systems, vol. 27, no. 8, p. 2550014, 2025.
- J. Piza-Londoño and **E. J. Andrade-Lotero**, "Leveraging semantic parsing using text embeddings and reinforcement learning," 2024 IEEE Latin American Conference on Computational Intelligence (LA-CCI), pp. 1–6, 2024.
- R. L. Goldstone, **E. J. Andrade-Lotero**, R. Hawkins, and M. Roberts, "The emergence of specialized roles within groups," Topics in Cognitive Science, vol. 16, pp. 257–281, 2023.
- **E. J. Andrade-Lotero**, J. A. Velasco, J. Ortiz-Duque, and R. L. Goldstone, "The division of linguistic labor for offloading conceptual understanding," Philosophical Transactions of the Royal Society B, vol. 378, no. 20210360, 2022.
- **E. J. Andrade-Lotero** and R. L. Goldstone, "Self-organized division of cognitive labor," PLOS ONE, vol. 16, no. 7, 2021.
- **E. J. Andrade-Lotero** and J. Ortiz-Duque, "A mis-conception of logic in its diffusion across non-mathematical environments," Linguistic and Philosophical Investigations, vol. 18, pp. 77–96, 2019.
- C. Álvarez, **E. J. Andrade-Lotero**, and V. Gauthier, "Modelos epidemiológicos en redes: Una presentación introductoria," Boletín de Matemáticas, vol. 22, no. 1, pp. 21–37, 2015.
- **E. J. Andrade-Lotero**, "Una gramática lógica para explicar algunos aspectos de la semántica del español," Boletín de Matemáticas, vol. 20, no. 1, pp. 21–50, 2013.
- **E. J. Andrade-Lotero** and C. Dutilh, "A contentious trinity: Levels of entailment in Brandom's pragmatic inferentialism,"

Philosophia, vol. 40, no. 1, pp. 41–53, 2012.

- **E. J. Andrade-Lotero** and C. Dutilh, "Validity, the squeezing argument and alternative semantic systems: The case of Aristotelian syllogistic," Journal of Philosophical Logic, vol. 41, no. 2, pp. 387–418, 2012.
- **E. J. Andrade-Lotero** and E. Becerra, "Establishing connections between Aristotle's natural deduction and first-order logic," History and Philosophy of Logic, vol. 29, no. 4, pp. 309–325, 2008.
- **E. J. Andrade-Lotero** and E. Becerra, "Corcoran's Aristotelian syllogistic as a subsystem of first-order logic," Revista Colombiana de Matemáticas, vol. 41, no. 1, pp. 67–80, 2007.

PUBLICATIONS: BOOKS AND TRANSLATIONS

- L. Gamut, Lógica, lenguaje y significado: lógica intensional y gramática lógica, **E. Andrade-Lotero** and C. M. (translators), Eds. Bogotá: Editorial Universidad del Rosario, 2010.
- **E. J. Andrade-Lotero**, P. Cubides, C. Márquez, E. J. Vargas, and D. Cancino, Lógica y Pensamiento Formal. Editorial Universidad del Rosario, 2008.

PUBLICATIONS: CONFERENCE PROCEEDINGS (EDITED)

- M. S. Dora Achouroti, **Edgar J. Andrade-Lotero**, Eds., Proceedings of the Amsterdam Graduate Philosophy Conference: Meaning and Truth, X-2010-01, University of Amsterdam, ILLC Publications, 2010.
- M. S. Dora Achouroti, **Edgar J. Andrade-Lotero**, Eds., Proceedings of the Graduate Philosophy Conference on Normativity, X-2009-02, University of Amsterdam, ILLC Publications, 2009.

PUBLICATIONS: CONFERENCE PAPERS

- **E. J. Andrade-Lotero** and R. L. Goldstone, "An empirical study of the El Farol Bar problem," Collective Intelligence, 2024.
- **E. J. Andrade-Lotero** and R. L. Goldstone, "Self-organized division of cognitive labor," Proceedings of the 41st Annual Conference of the Cognitive Science Society, 2019, pp. 91–97.
- **E. J. Andrade-Lotero** and R. L. Goldstone, "Self-organized division of cognitive labor," Collective Intelligence, 2019.
- **E. J. Andrade-Lotero** and J. Ortiz, "The proper approach to the division of linguistic labor," Sixth International Conference on Philosophy of Language and Linguistics, 2019.
- **E. J. Andrade-Lotero**, "Translations and structures: Some reflexions about teaching the semantics of first-order logic," Philogica IV, 2016.
- **E. J. Andrade-Lotero**, "Explicating compositionality," Philogica III, 2014.
- **E. J. Andrade-Lotero**, "On truth conditions that depend on performatives," Third International Conference on Philosophy of Language and Linguistics, 2013.
- **E. J. Andrade-Lotero**, "El principio de composicionalidad y la competencia lingüística," IV Congreso Colombiano de Filosofía, 2012.
- **E. J. Andrade-Lotero**, "Compositionality from a 'use-theoretic' perspective," 32nd International Wittgenstein Symposium, 2009.
- **E. J. Andrade-Lotero** and C. Dutilh, "A reconstruction of the hierarchical relations between incompatibility entailment, committive and permissive consequence," Proceedings of the Workshop on Bob Brandom's Recent Philosophy of Language: Towards an Analytic Pragmatism, 2009, pp. 90–98.

STUDENTS SUPERVISED

- J. V. Tamayo-Ramírez, Marco general para diseñar y evaluar sistemas de tutoría inteligentes: una propuesta para aprovechar grandes modelos de lenguaje de manera rigurosa, Masters in Engineering, Science and Technology, 2026.
- J. P. Rivera Burgos, Visibilizar lo invisibilizado: herramienta de procesamiento de lenguaje natural para sistematizar noticias de homicidios de población LGBTIQ+, Masters in Applied Mathematics and Computer Science, 2025.
- C. A. Gómez Vargas, Integración del aprendizaje por refuerzo en parsers semánticos para la deducción lógica en lenguaje natural, Masters in Applied Mathematics and Computer Science, 2025.
- S. Carrerá Martínez, Política de puntos y líneas: aplicación de teoría de grafos al comportamiento de los partidos políticos colombianos en el Congreso de la República, Masters in Applied Mathematics and Computer Science, 2025.
- E. J. López-López and S. Gaitán-Rubio, Modelación de precios por medio del juego de las minorías, Masters in Applied Mathematics and Computer Science, 2024.
- E. Muñoz, Gestión activa de portafolios de cripto activos utilizando técnicas de aprendizaje por refuerzo, Masters in Applied Mathematics and Computer Science, 2024.
- J. Piza-Londoño, Aprendizaje por refuerzo de un parser semántico óptimo en DRT, Masters in Applied Mathematics and Com-

puter Science, 2024.

- J. V. Tamayo-Ramírez, G. A. N. Cárcamo, and R. D. Rico-González, Automatización de procesos de cualificación en IQAP mediante chatbot de IA para ingreso a universidades internacionales, Masters in Applied Mathematics and Computer Science, 2024.
- J. A. Velasco, Hacia una cartografía del significado: Una exploración por los roles del sentido en el Tractatus-Logico-Philosophicus, Bachelors in Philosophy, 2019.
- M. P. Gómez, La incompletitud de los fundamentos de las matemáticas, Bachelors in Philosophy, 2018.
- M. G. Algarra, Inconmensurabilidad e interpretación: Una propuesta de comparación entre teorías científicas, Bachelors in Philosophy, 2007.

COURSES TAUGHT

- 2021–2024 (graduate): Reinforcement Learning; Artificial Intelligence: Problem Representation and Solving; Machine Learning Applications — Universidad del Rosario.
- 2017–2024 (undergraduate): Logic for Computer Science; Logic, Number Theory and Sets; Logic and Discrete Mathematics; Theory of Computation; Linear Algebra; Artificial Intelligence: Problem Representation and Solving; Numerical Analysis and Scientific Computing — Universidad del Rosario.
- 2007–2016 (undergraduate): Integral Calculus; Probability; Inferential Statistics; Philosophy of Language; Logic; Logical-Philosophical Models of Natural Language; Pragmatics; Formal Methods; Processes and Algorithms; Epidemiology; Scientific LaTeX — Universidad del Rosario.
- 2006: History of Mathematics; Theory of Computation — Universidad Sergio Arboleda.
- 2003: Mathematical Logic — Universidad Sergio Arboleda.
- 2002: Logic for Psychologists — Universidad Nacional de Colombia.

CERTIFICATIONS

- Reinforcement Learning Specialization, Coursera
- Generative AI: Prompt Engineering Basics, IBM
- Introduction to Artificial Intelligence (AI), IBM
- Introduction to Deep Learning & Neural Networks with Keras, IBM
- Generative AI Engineering, Coursera
- Generative AI with Large Language Models, Coursera
- Machine Learning Engineering for Production, Coursera
- Deep Learning, Coursera
- Reinforcement Learning for Trading, Cognitive Class
- Machine Learning with Python (with Honors), IBM
- Python for Data Science, AI & Development, IBM

SKILLS

Programming Languages: Python, PyTorch, R, JavaScript, LaTeX

Spoken Languages: English (C1, TOEFL), French (A2, DELF)

AWARDS & HONORS

- 2022: Visiting Scholar Grant, Universidad del Rosario
- 2020: Research Seedbed Grant, Universidad del Rosario
- 2018: Visiting Scholar Grant, Universidad del Rosario
- 2018: Seed Capital Grant, Universidad del Rosario
- 2004: Huygens Scholarship for Outstanding Foreign Students, Netherlands Organization for International Cooperation in Higher Education